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```
clc;
clear;
close all;
```

Battery Safety Parameters

```
% Battery operating values
voltage = 52;          % Battery voltage (V)
current = 40;          % Battery current (A)
temperature = 35;     % Battery temperature (deg C)
SOC = 85;              % State of Charge (%)
```

Safety Limits

```
max_voltage = 54;      % Maximum allowable voltage (V)
min_voltage = 39;      % Minimum allowable voltage (V)

max_current = 50;      % Maximum allowable current (A)

max_temperature = 60;  % Maximum allowable temperature (deg C)

max_SOC = 100;         % Maximum SOC (%)
min_SOC = 0;           % Minimum SOC (%)
```

Safety Checks

```
voltage_status = true;
current_status = true;
temperature_status = true;
SOC_status = true;

% Voltage check
if voltage > max_voltage
    fprintf('WARNING: Over-voltage detected!\n');
    voltage_status = false;

elseif voltage < min_voltage
    fprintf('WARNING: Under-voltage detected!\n');
    voltage_status = false;

else
```

```
    fprintf('Voltage Status: SAFE\n');
end

% Current check
if abs(current) > max_current
    fprintf('WARNING: Over-current detected!\n');
    current_status = false;

else
    fprintf('Current Status: SAFE\n');
end

% Temperature check
if temperature > max_temperature
    fprintf('WARNING: Over-temperature detected!\n');
    temperature_status = false;

else
    fprintf('Temperature Status: SAFE\n');
end

% SOC check
if SOC > max_SOC
    fprintf('WARNING: Overcharge detected!\n');
    SOC_status = false;

elseif SOC < min_SOC
    fprintf('WARNING: SOC below allowable limit!\n');
    SOC_status = false;

else
    fprintf('SOC Status: SAFE\n');
end
```

```
Voltage Status: SAFE
Current Status: SAFE
Temperature Status: SAFE
SOC Status: SAFE
```

Overall Battery Safety Status

```
if voltage_status && current_status && temperature_status && SOC_status
    fprintf('\nOverall Battery Status: SAFE\n');
else
    fprintf('\nOverall Battery Status: UNSAFE\n');
end
```

```

Overall Battery Status: SAFE
```

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